

```

clear;
clc;
close all;

%% Load audio files

[noisy, Fs1] = audioread('mixed_gunshot_speech.wav');
[reference, Fs2] = audioread('gunshot_noise_only.wav');

%% Display sampling rates

fprintf('Primary sample rate   : %d Hz\n', Fs1);

```

Primary sample rate : 16000 Hz

```

fprintf('Reference sample rate : %d Hz\n', Fs2);

```

Reference sample rate : 16000 Hz

```

%% Make sure both signals have the same length

N = min(length(noisy), length(reference));

noisy = noisy(1:N);
reference = reference(1:N);

%% Create time vector

t = (0:N-1) / Fs1;

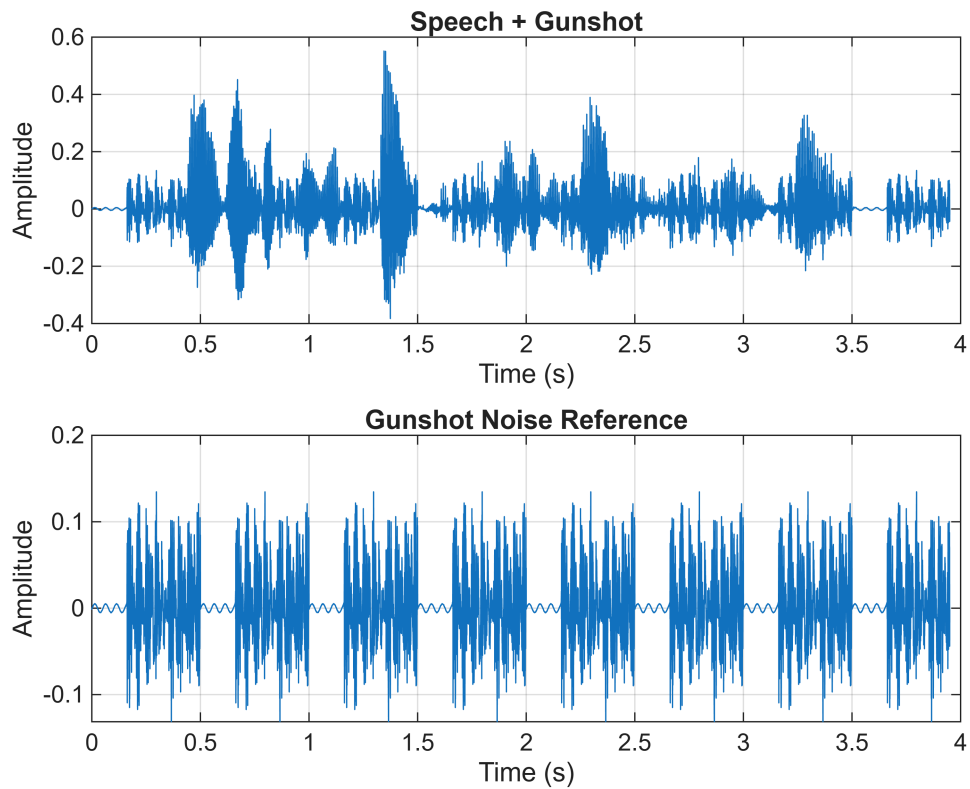
%% Plot Speech + Noise and Noise Reference

figure;

subplot(2,1,1);
plot(t, noisy);
title('Speech + Gunshot');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(2,1,2);
plot(t, reference);
title('Gunshot Noise Reference');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

```



```

%% Estimate the actual noise
% Used only as ground truth for comparison
%
% Since we already have the actual gunshot-only recording,
% it is our ground-truth noise.

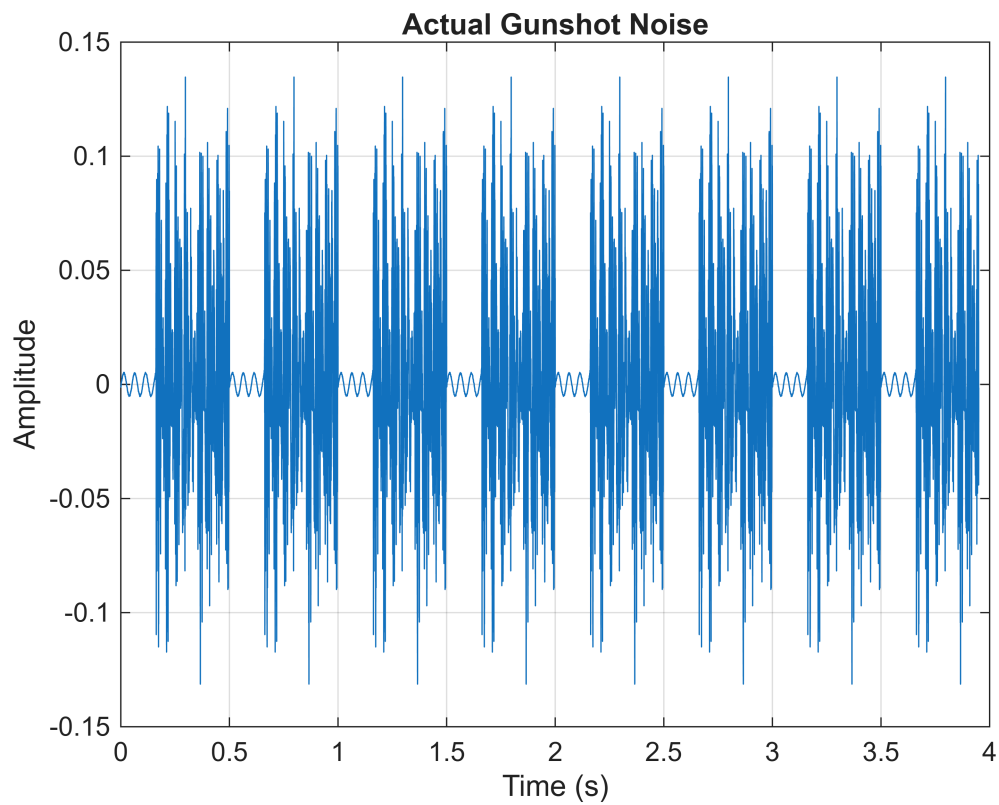
noise = reference;

%% Plot the actual noise

figure;

plot(t, noise);
title('Actual Gunshot Noise');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

```



```
%% STFT of Speech + Gunshot
```

```
figure;
```

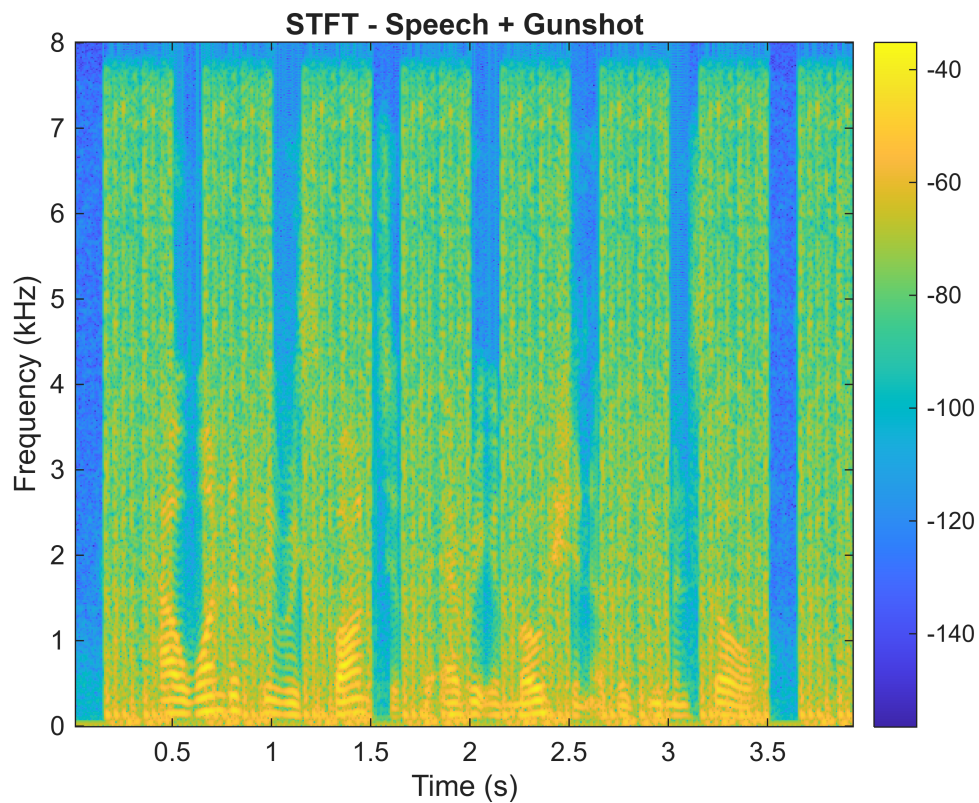
```
spectrogram(noisy, hamming(512), 384, 1024, Fs1, 'yaxis');
```

```
title('STFT - Speech + Gunshot');
```

```
xlabel('Time (s)');
```

```
ylabel('Frequency (kHz)');
```

```
colorbar;
```



```
%% NLMS Adaptive Filter
%
% Primary    = Speech + Gunshot
% Reference = Gunshot Only
%
% NLMS estimates the gunshot/noise component.
% The residual is the enhanced speech.
```

```
primary = noisy;
reference = reference;
```

```
% NLMS parameters
```

```
filterLength = 1;
mu = 0.05;
epsilon = 1e-4;
```

```
% Initialize filter coefficients
```

```
w = zeros(filterLength, 1);
```

```
% Initialize output signals
```

```
estimatedNoise = zeros(N, 1);
enhancedSpeech = zeros(N, 1);
```

```
%% NLMS processing
```

```

for n = filterLength:N

    % Reference signal vector
    x = reference(n:-1:n-filterLength+1);

    % Estimate noise component
    estimatedNoise(n) = w' * x;

    % Residual = enhanced speech
    enhancedSpeech(n) = primary(n) - estimatedNoise(n);

    % Update NLMS filter coefficients
    w = w + (mu / (epsilon + x' * x)) ...
        * enhancedSpeech(n) * x;

end

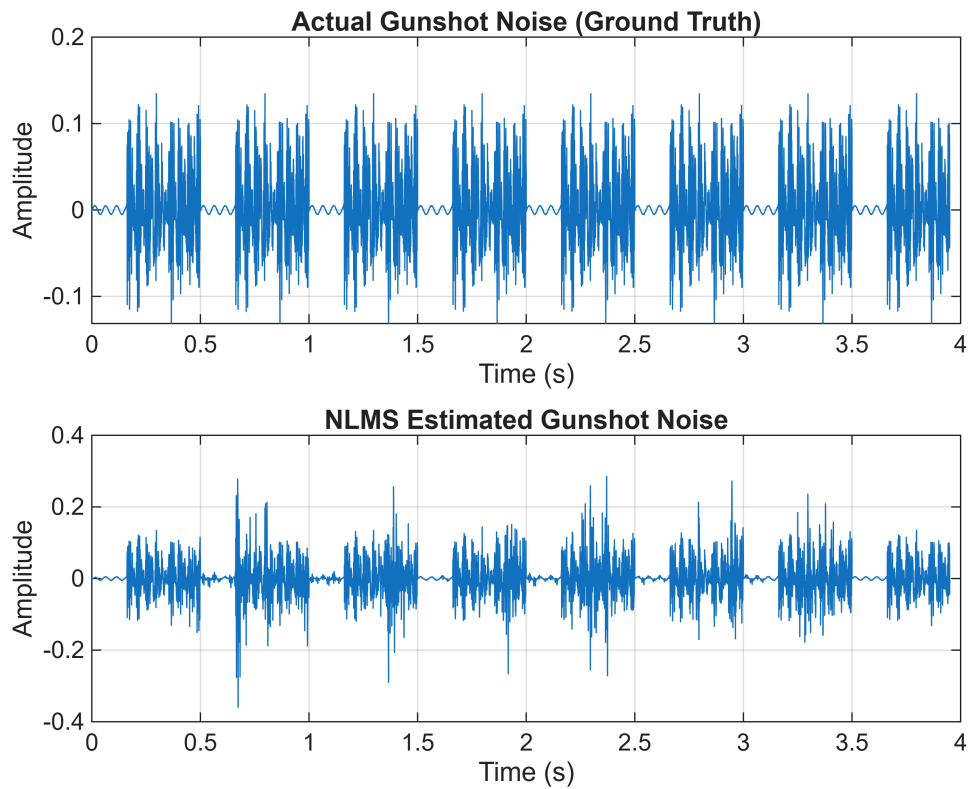
%% Compare Actual Noise and NLMS Estimated Noise

figure;

subplot(2,1,1);
plot(t, noise);
title('Actual Gunshot Noise (Ground Truth)');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(2,1,2);
plot(t, estimatedNoise);
title('NLMS Estimated Gunshot Noise');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

```



```
%% Compare Original and Enhanced Speech
```

```
figure;
```

```
subplot(2,1,1);
```

```
plot(t, primary);
```

```
title('Original - Speech + Gunshot');
```

```
xlabel('Time (s)');
```

```
ylabel('Amplitude');
```

```
grid on;
```

```
subplot(2,1,2);
```

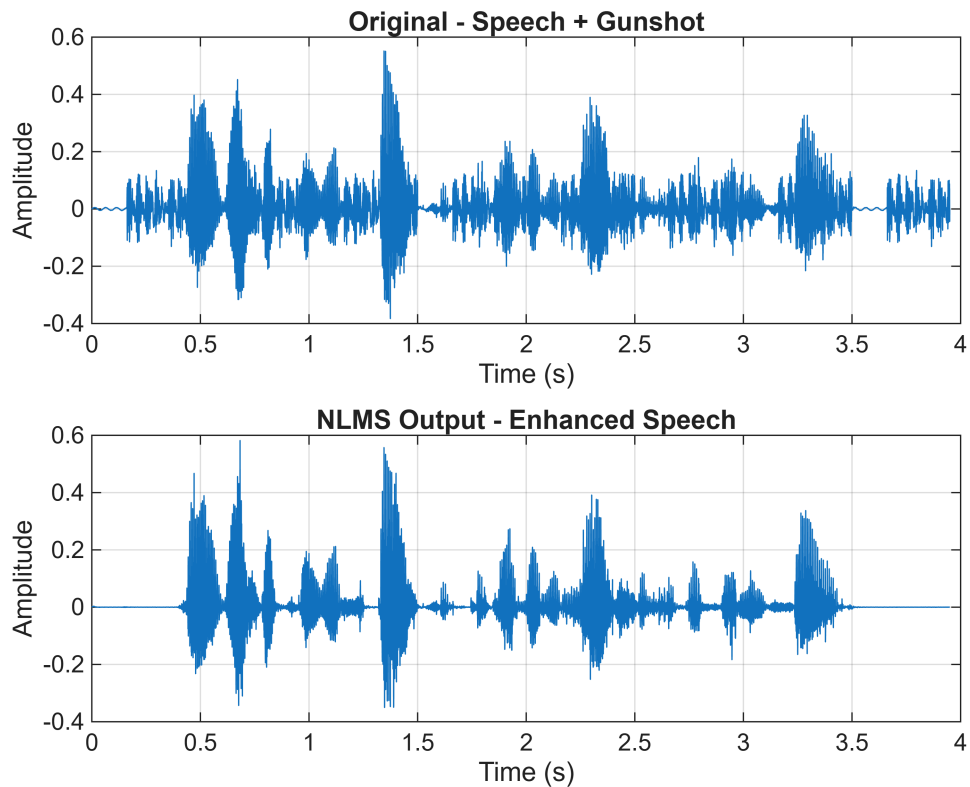
```
plot(t, enhancedSpeech);
```

```
title('NLMS Output - Enhanced Speech');
```

```
xlabel('Time (s)');
```

```
ylabel('Amplitude');
```

```
grid on;
```



```
%% Performance Evaluation
```

```
% Remove initial filter samples from comparison
```

```
valid = filterLength:N;
```

```
actual = noise(valid);
```

```
estimated = estimatedNoise(valid);
```

```
% Mean Squared Error
```

```
MSE = mean((actual - estimated).^2);
```

```
% Root Mean Squared Error
```

```
RMSE = sqrt(MSE);
```

```
% Correlation coefficient
```

```
R = corrcoef(actual, estimated);
```

```
correlation = R(1,2);
```

```
% Display results
```

```
fprintf('\n--- NLMS Performance ---\n');
```

```
--- NLMS Performance ---
```

```
fprintf('MSE : %.6f\n', MSE);
```

MSE : 0.000298

```
fprintf('RMSE : %.6f\n', RMSE);
```

RMSE : 0.017259

```
fprintf('Correlation : %.4f\n', correlation);
```

Correlation : 0.8451

```
%% Save NLMS Estimated Noise
```

```
audiowrite('NLMS_estimated_noise.wav', estimatedNoise, Fs1);
```

```
fprintf('NLMS estimated noise saved as NLMS_estimated_noise.wav\n');
```

NLMS estimated noise saved as NLMS_estimated_noise.wav

```
%% Save Enhanced Speech
```

```
audiowrite('NLMS_enhanced_speech.wav', enhancedSpeech, Fs1);
```

```
fprintf('NLMS enhanced speech saved as NLMS_enhanced_speech.wav\n');
```

NLMS enhanced speech saved as NLMS_enhanced_speech.wav